ROUND 2 — THE BUILD SUBMISSION

PRODUCTION HACKATHON SUBMISSION

BRANDSENTRY AI

Explainable, evidence-grounded brand governance before marketing copy goes live.

A deployed, production-verified RAG compliance review system that evaluates marketing claims against approved brand guidelines and regulatory playbooks. Features deterministic sliding-window PyMuPDF chunking, ONNX MiniLM vector embeddings, ChromaDB semantic retrieval, 3-way verdicts (SUPPORTED, CONTRADICTED, INSUFFICIENT_EVIDENCE), character-exact citation grounding, and an explainable 7-category semantic drift detection engine.

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Nexora AI

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SUBMISSION DATE
24 October 2026

ROUND 2 DEADLINE
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1 Executive Summary

Overview of the operational Round 2 platform, delivered capabilities, and public deployment

Transforming Policy Documents into Deterministic Pre-Publication Safeguards

BrandSentry AI turns approved brand guidelines into a practical pre-publication review step. A user uploads a guideline document (PDF or TXT), enters marketing copy, and receives deterministic verdicts grounded directly in retrieved source passages. The application explicitly distinguishes a claim supported by verified evidence from one contradicted by policy, and from one for which the available sources are insufficient — preventing fabricated approvals.

What Is Working in Production

• Guideline Ingestion Engine

Multipart upload supporting PDF and TXT with PyMuPDF digital extraction, page boundary preservation, and automatic fallback rejection for scanned non-digital PDFs.

• Evidence-Grounded Review

Dense vector retrieval using local ONNX `all-MiniLM-L6-v2` embeddings and persistent ChromaDB with character-exact citation substring validation.

• 3-Way Verdict Classification

Conservative, explainable triage: `SUPPORTED`, `CONTRADICTED`, and `INSUFFICIENT_EVIDENCE` to eliminate hallucinated approvals.

• Deterministic Safeguards

High-risk prohibited phrase screening (e.g. absolute guarantees, unverified superlatives) and conservative handling of missing evidence.

• Semantic Drift Detection

Deterministic analysis across 7 linguistic categories with severity rating (LOW/MED/HIGH), evidence phrase diffing, and conservative compliant rewrites.

• Accessible Production Cloud

Dockerized FastAPI backend and React + TypeScript static frontend deployed independently on Render with automated HTTPS and health check monitoring.

Live Production Endpoints

Service	Public URL	Status & Verification
Web Application UI	`https://brandsentry-web.onrender.com`	LIVE • 200 OK React + TypeScript + Vite CDN
Backend REST API	`https://brandsentry-api.onrender.com`	LIVE • 200 OK FastAPI on Docker / Uvicorn
Interactive API Docs	`https://brandsentry-api.onrender.com/docs`	SWAGGER UI Interactive schema testing
API Health Check	`https://brandsentry-api.onrender.com/health`	HEALTHY Returns uptime, version, and status
Source Code Repository	`https://github.com/KPCHIRAGGUPTHA/BRAND_SENTRY_AI`	PUBLIC GIT Complete commit & test history

Important Scope & Responsibility Notice

This application is an AI-assisted decision-support prototype designed to enhance human reviewer accuracy and velocity. It does not constitute legal certification or guaranteed detection of all regulatory violations. Final publication review remains a human responsibility.

2 Problem Statement & Product Goal

Market challenges in pre-publication content governance and the BrandSentry AI design principles

The Real-World Governance Challenge

Marketing copy is generated at unprecedented velocity by decentralized internal teams, external creative agencies, and generative AI tools. However, corporate brand guidelines and legal policies are dense, constantly updated, and trapped inside long PDF manuals. As a result:

- **Unsubstantiated and Outdated Claims:** Marketing statements exaggerate product capabilities or quote deprecated statistics without citations.
- **Prohibited Guarantees:** Copywriters frequently use banned absolute claims (e.g., "100% risk-free", "guaranteed cure", "fastest on earth") that trigger regulatory action.
- **Semantic Drift:** Qualified internal facts ("may assist in recovery") become absolute promotional promises ("guarantees complete recovery") through subtle paraphrase.
- **Review Bottlenecks:** Human legal and brand review takes days or weeks, creating friction between marketing speed and compliance rigor.

Product Principles Retained from Round 1

1. Evidence Before Explanation

No claim is marked compliant without direct character-exact substring evidence from an uploaded, authorized brand guideline. Unsubstantiated claims must never receive silent approval.

2. Missing Evidence ≠ Falsity

The system strictly separates `INSUFFICIENT_EVIDENCE` from `CONTRADICTED`. If a claim is not addressed in policy, the engine reports missing evidence rather than guessing.

3. Explain and Correct

Beyond issuing a verdict, the system produces actionable explanations and conservative suggested rewrites that preserve promotional tone while eliminating compliance risk.

4. Human-in-the-Loop Oversight

BrandSentry AI is designed as reviewer-augmentation software. It exposes full audit evidence, confidence scores, and reasoning to empower human sign-off.

Primary User Journey Implemented in Round 2

1. Upload Guidelines
PDF/TXT drag & drop; PyMuPDF text chunking

→

2. Input Copy
Multi-sentence marketing copy; atomic claim parsing

→

3. RAG & Drift Triage
MiniLM vector retrieval; 7-drift checks & rules

→

4. Review Findings
Verdicts, citations, drawer diffs & rewrites

3 Progress from Round 1 to Round 2

Transparent comparison between initial proposal promises, delivered build features, and deferred roadmap

Round 1 Proposal Plan	Round 2 Implementation Status	Technical Evidence & Verification Notes
Guideline Document Ingestion	DEPLOYED	PyMuPDF digital extraction, page-aware chunking (~400 tokens / 50 overlap), MIME/size validation. Tested in `test_extraction.py` & `test_api_guidelines.py`.
Vector-Based Evidence Retrieval	DEPLOYED	Pure local ONNX `all-MiniLM-L6-v2` 384-dim embeddings + ChromaDB vector store. Tested across `test_embeddings.py` & `test_vector_store.py`.
Three-Way Claim Verdicts	DEPLOYED	Deterministic classification into `SUPPORTED`, `CONTRADICTED`, and `INSUFFICIENT_EVIDENCE`. Covered by 52 unit tests in `test_verdict_engine.py`.
Citation Grounding & Page Tracking	DEPLOYED	Character-exact substring verification against raw source text with page numbers. Hallucinated quotes strictly rejected. Verified in `TestCitationValidation`.
Prohibited Phrase Pre-Screening	DEPLOYED	Regex and keyword engine flagging absolute guarantees, false health claims, and unhedged superlatives. Verified in `TestProhibitedPhraseRules`.
Semantic Drift Detection	DEPLOYED & TESTED	Added during Round 2: 7 heuristic drift categories with diffing, severity rating, and suggested fixes. 15 dedicated tests in `test_semantic_drift.py`.
Review Queue, Comments & Approvals	DEFERRED	Deferred to Round 3 to focus engineering resources on building a robust, high-reliability claim verification engine first.
Multi-Tenant Auth & Enterprise RBAC	DEFERRED	Public demo mode enabled for hackathon judging convenience without requiring login credentials or tenant provisioning.
Multimodal Visual & Logo Compliance	ROADMAP (R3)	Text-first PDF/TXT focus delivered. Logo, color, and image disclaimer verification is slated as the primary Round 3 expansion.

Engineering Rationale: The Focused Vertical Slice

In Round 1, an expansive brand-governance suite was outlined. For Round 2 (The Build), we intentionally prioritized a **bulletproof, demonstrable vertical slice**: taking real PDFs, reliably extracting text, generating local embeddings without API costs, finding citations, evaluating claims with zero hallucinated approvals, and detecting semantic drift. This delivers an honest, operational tool that judges can test immediately.

4 System Architecture & Data Flow

End-to-end component breakdown from presentation layer down to storage and embedding models

Layer	Component & Technology	Architectural Role & Responsibilities
Presentation Layer	React 18 + TypeScript + Vite + Tailwind/CSS	Drag-and-drop document uploader, marketing copy review form, interactive verdict breakdown cards, expandable evidence drawers, and semantic drift diff viewer.
API & Routing	FastAPI (Python 3.11/3.14) + Uvicorn	RESTful endpoints (`/api/guidelines/upload`, `/api/review`, `/health`), Pydantic v2 request/response validation, CORS header management, and exception handling.
Extraction Layer	PyMuPDF 2.0 (`fitz`)	Extracts digital text while retaining physical page boundaries; detects scanned image-only PDFs and returns clean HTTP 422 diagnostic responses.
Vector Pipeline	ONNX `all-MiniLM-L6-v2` + ChromaDB	Runs local CPU inference to produce 384-dimensional unit L2 normalized dense embeddings. Persists vectors to disk for cosine distance retrieval.
Verdict Engine	Multi-tier Rule Engine + Heuristics	Screens prohibited phrases, applies distance thresholds, enforces character-exact substring citation validation, and coordinates optional LLM reasoning.
Metadata Store	SQLite (`data/documents.db`)	Tracks uploaded document UUIDs, filenames, total chunks, character lengths, and processing timestamps.
Model Agnosticism	Pluggable Fallback Architecture	Can run completely standalone on pure deterministic CPU heuristics, or connect to server-side Gemini/OpenAI API keys for nuanced rewrite generation.

Runtime Request Execution Pipeline

```
[Marketing Copy] → FastAPI /api/review → Claim Extractor (Splits into Atomic Propositions) ↳ Prohibited Phrase Scanner (Regex match → Direct CONTRADICTED verdict if found) ↳ ONNX MiniLM Embedding (384-dim vector generated via local CPU) ↳ ChromaDB Cosine Query (Retrieves top-k candidate guideline chunks) ↳ Citation Substring Verifier (Rejects quotes not found verbatim in source) ↳ Semantic Drift Engine (Compares claim vs evidence across 7 categories) ↳ Pydantic Response Serializer → React Review Dashboard
```

Data Privacy & Ephemeral Cloud Notice

The public demo is deployed on Render's free tier. All guideline text and embeddings are stored on ephemeral container disks without tracking personal data. Because free-tier containers restart after idle periods, demo users can re-upload guidelines at any time using the one-click sample guideline button in the web UI.

5 Round 2 Advancement: Semantic Drift Detection

Detecting subtle, dangerous shifts in meaning between internal facts and external marketing claims

Why Vector Similarity Alone Is Insufficient

Standard dense vector retrieval measures topical proximity, not factual precision. A claim can have **95% vector similarity** to a guideline passage while reversing its truth value or removing critical legal qualifications. BrandSentry AI introduces a deterministic linguistic drift detector that evaluates claim-evidence pairs across 7 distinct categories.

Drift Category	What the Engine Evaluates	Illustrative Linguistic Example
CERTAINTY_SHIFT	Modality change from possibility or probability to an absolute certainty.	*"may improve efficiency"* → **"always improves efficiency"**
SCOPE_EXPANSION	Broadening audience or eligibility from a qualified cohort to everyone.	*"registered enterprise users"* → **"all customers"**
NUMERICAL_EXAGGERATION	Claimed numerical value, metric, or percentage exceeds evidence figure.	*"up to 15% faster"* → **"85% faster"**
CONDITION_OMISSION	Removing prerequisites, dependencies, or required caveats.	*"if unused within 30 days"* → **"within 30 days"**
TEMPORAL_SHIFT	Altering timing, duration, latency, or frequency parameters.	*"usually ships within 48h"* → **"instant dispatch"**
CAUSALITY_SHIFT	Upgrading an observed correlation into an unproven direct cause.	*"is associated with lower churn"* → **"eliminates customer churn"**
COMPARISON_SHIFT	Transforming a qualified comparative into an unsubstantiated superlative.	*"competitive performance"* → **"fastest solution in the world"**

What the UI Presents to Reviewers

- **Severity Rating:** `LOW`, `MEDIUM`, or `HIGH`.
- **Phrase-Level Comparison:** Specific claim phrase vs source phrase.
- **Conservative Suggested Rewrite:** Restores compliant phrasing while maintaining promotional punch.

Deterministic Guardrails

- **Zero Fabrication:** Drift engine never invents evidence citations.
- **Topical Overlap Filter:** Rejects false flags when sentences discuss unrelated subjects.
- **Preserves Verdict Semantics:** A drift flag supplements, but never blindly overturns, verified evidence.

6 Demonstration Scenarios & Expected Behaviour

Test scenarios for judges to verify deterministic safeguards and semantic drift on the live web deployment

Test Scenario	Uploaded Guideline Passage	Submitted Marketing Claim	Expected Engine Behaviour
1. Evidence-Supported Claim	*"Marketing materials must include traceable evidence sources for every material factual claim."*	"Our marketing materials must include traceable evidence sources for every material factual claim."	**SUPPORTED** Returns character-exact guideline quote, page number, and high confidence.
2. Unsupported Guarantee	*"Approved guidelines strictly prohibit absolute safety and performance guarantees without certified testing."*	"Our product is 100% guaranteed to deliver perfect results."	**CONTRADICTED** Flags prohibited term ("100% guaranteed"), returns policy citation, and suggests compliant rewrite.
3. Missing Evidence (No Hallucination)	*(No guideline passage mentioning customer productivity metrics or percentages exists in index).*	"Our enterprise platform increases customer productivity by 85%."	**INSUFFICIENT_EVIDENCE** Correctly reports lack of evidence; strictly avoids hallucinating an approval.
4. Certainty Drift Detection	*"Our platform may improve operational productivity under suitable conditions."*	"Our platform always improves operational productivity."	**DRIFT DETECTED** Flags `CERTAINTY_SHIFT` with `HIGH` severity; highlights "may" vs "always" diff.
5. Condition Omission	*"Full refunds are available within 30 days when the product is unused and undamaged."*	"Full refunds are available within 30 days."	**DRIFT DETECTED** Flags `CONDITION_OMISSION`; alerts reviewer that required condition was dropped.

Recommended Judge Demonstration Walkthrough (3 Minutes)

1. **Open the Web Dashboard:** Navigate to https://brandsentry-web.onrender.com.
2. **Load Sample Guideline:** Click *"Load Sample Marketing Compliance Guidelines"* (or upload `BrandSentry_Sample_Marketing_Compliance_Guidelines.txt`). Confirm the document status card displays chunk counts.
3. **Run a Supported Claim:** Paste Scenario 1. Observe the green `SUPPORTED` card and expandable citation drawer showing page boundaries.
4. **Run an Absolute Guarantee:** Paste Scenario 2. Observe the red `CONTRADICTED` card, prohibited phrase flag, and suggested correction.
5. **Verify Zero Hallucination:** Paste Scenario 3. Observe the amber `INSUFFICIENT_EVIDENCE` tag and absence of fabricated quotes.
6. **Inspect Semantic Drift:** Paste Scenario 4. Observe the Semantic Drift Analysis section showing `CERTAINTY_SHIFT`.

7 Completed Features & Current Boundaries

Comprehensive audit of delivered features, automated test coverage, and deferred capabilities

166	0	0	16
BACKEND TESTS PASSING	FAILED TESTS (100% PASS)	TYPESCRIPT ERRORS	DEPLOYED FEATURES

Feature Module	Verified Status	Current Implementation & Verification Evidence
Web UI & REST API	DEPLOYED	FastAPI container & Vite static app running live on Render with automated HTTPS and CORS.
PDF & TXT Ingestion	DEPLOYED	PyMuPDF digital extraction, page-aware chunking, and scanned image-only PDF rejection (HTTP 422).
Dense Vector Retrieval	DEPLOYED	Local ONNX MiniLM 384-dimensional embeddings + persistent ChromaDB with cosine distance scoring.
Three-Way Verdicts	DEPLOYED	Deterministic classification into `SUPPORTED`, `CONTRADICTED`, and `INSUFFICIENT_EVIDENCE`.
Prohibited Phrase Screening	DEPLOYED	Pre-screening engine blocking absolute guarantees, medical cure claims, and unhedged superlatives.
Semantic Drift Engine	DEPLOYED & TESTED	7 heuristic categories, phrase-level comparison diffs, severity tags, and conservative rewrite suggestions.
Automated Regression Suite	166 PASSING	Full suite of 166 pytest tests covering health, extraction, chunking, embeddings, verdicts, and drift.
Frontend Production Build	0 TS ERRORS	`tsc && vite build` compiles 1,593 modules cleanly into production assets without warnings.

Explicit Boundaries & Deferred Scope

To maintain absolute engineering integrity, the following capabilities are **not claimed as completed** in Round 2:

- **Enterprise Multi-Tenancy & RBAC:** JWT user auth and isolated multi-client databases are deferred.
- **Approval Workflows & Audit Logs:** Multi-user comment threads, reviewer sign-offs, and version diffing remain Round 3 goals.
- **Sparse BM25 + Dense Hybrid Retrieval:** Retrieval is currently purely dense vector search; hybrid rank fusion is deferred.
- **Multimodal Creative Verification:** Image-based logo placement, typography rules, and video audits remain roadmap items.

8 Technology Stack & Verification Audit

Comprehensive technology choices, architecture justification, and empirical verification results

Layer / Area	Technology Selected	Engineering Justification & Purpose
Frontend Framework	React 18, TypeScript, Vite	Component modularity, type-safe API client, sub-second HMR development, and compact production bundle.
Backend Framework	Python, FastAPI, Uvicorn	High-performance asynchronous REST endpoints, automatic OpenAPI / Swagger generation, Pydantic v2 schemas.
PDF Processing	PyMuPDF 2.0 (`pymupdf`)	Industry-standard C-speed text extraction that strictly tracks physical page numbers for audit citations.
Embedding Inference	ONNX `all-MiniLM-L6-v2`	Self-contained CPU inference generating 384-dim normalized vectors without external API calls or latency costs.
Vector Database	ChromaDB (Local persistent)	Fast cosine distance index with document-level metadata filtering and idempotent upsert support.
Document Registry	SQLite (`documents.db`)	Embedded SQL metadata database recording uploaded guideline UUIDs, chunk counts, and file sizes.
Cloud Infrastructure	Render Container & Static CDN	Fully automated CI/CD deployment triggered directly on pushes to GitHub `main` branch.
Testing Frameworks	Pytest, TypeScript Compiler	Automated unit, integration, and regression testing covering all API endpoints and verification rules.

Empirical Verification Results (Audited 24 October 2026)

• Backend Automated Suite: 166 Passed • 0 Failed • 4 Warnings

Executed locally via `pytest -v` in 13.72 seconds. Tests verify health endpoints (5), PyMuPDF extraction (8), sliding-window chunking (7), ONNX embeddings (11), vector distance semantics (16), ChromaDB store (4), guideline APIs (15), claim extraction (16), review schemas (17), 3-way verdict engine (52), and semantic drift engine (15).

• Frontend Typecheck & Production Build: 0 Errors

Executed via `npm run build` (invoking `tsc && vite build`). Successfully transformed 1,593 modules into production assets with zero TypeScript compilation warnings or errors.

9 Changes from Plan & Known Limitations

Intentional architectural scope adaptations and transparent mitigation strategies

Architectural Scope Adaptations Made During Round 2

Vertical Slice Prioritization

Narrowed the MVP scope to claim-level verification against uploaded guidelines rather than building an incomplete multi-role project management suite.

Added Semantic Drift Detection

Recognizing that vector similarity alone misses subtle qualification drops, we designed and shipped a deterministic 7-category drift engine in Round 2.

Simplified Workflow Lifecycle

Replaced multi-stage human approval state machines with an immediate, interactive reviewer dashboard featuring expandable evidence cards and instant diffs.

Pragmatic Cloud Constraints

Used embedded SQLite and local ChromaDB instead of external managed PostgreSQL and Redis clusters, ensuring complete zero-cost portability on Render free tier.

Known System Limitations & Mitigations

Limitation	Potential Failure Mode	Implemented Safeguard / Mitigation
Heuristic Drift Rules	May miss highly nuanced domain idioms or novel indirect paraphrases.	UI presents verbatim source passage; flags uncertainty and keeps human reviewer in control.
Vector Retrieval Misses	Guideline passage may not appear in top-k if vocabulary differs drastically.	Never claims a missing passage means the claim is false; safely returns `INSUFFICIENT_EVIDENCE`.
Ephemeral Free Hosting	Render free containers restart on idle, clearing local uploaded documents.	One-click sample guideline loader in UI enables instant re-hydration for demo sessions.
Scanned Document PDFs	PyMuPDF cannot read text from rasterized image scans without OCR.	HTTP 422 diagnostic rejection informs user document contains no extractable digital text.
No Legal Certification	AI outputs cannot replace licensed legal or regulatory compliance review.	Explicit disclaimers in UI; verdicts formatted strictly as decision-support suggestions.

10 Three Major Improvements Planned for Round 3

Substantial architectural upgrades targeting evidence quality, policy governance, and creative coverage

Improvement Area	What Will Be Engineered	Impact, Rationale & Validation Plan
1. Hybrid Evidence Retrieval & Evaluation Benchmark	Combine dense ChromaDB vector search with sparse BM25 lexical keyword retrieval via Reciprocal Rank Fusion (RRF). Develop a standardized benchmark dataset of 150+ labeled claims with ground-truth evidence spans.	Eliminates keyword misses for exact regulatory article citations (e.g. "Rule 24.b"). Empirically measures precision, recall, and Top-K retrieval hit rate against published baselines.
2. Auditable Policy Governance & Versioned Workflow	Introduce immutable guideline version snapshots (v1.0 → v1.1), diff visualizers, reviewer status state machines (Draft → Reviewed → Approved), and cryptographically signed PDF compliance audit export certificates.	Transforms BrandSentry AI from a single-user checker into an enterprise-grade governance platform. Validated through end-to-end review lifecycle tests.
3. Multimodal Creative & Visual Brand Compliance	Extend ingestion to image and banner assets (PNG/JPEG). Implement Tesseract/Vision OCR to extract text overlays, evaluate color palette adherence against brand hex tokens, and detect logo clear-space violations.	Addresses the #1 real-world marketing need: verifying complete creative assets, not just isolated copy. Validated on a benchmark of 50 sample marketing ad creatives.

Round 3 Quantitative Success Criteria

Retrieval Precision

Achieve ≥92% Top-3 evidence hit rate on our labeled regulatory benchmark dataset, with documented edge cases.

Traceable Governance

Generate 100% reproducible audit reports linking every approved marketing asset to the exact guideline commit hash.

Multimodal Coverage

Accurately parse text overlays and brand palette hex codes from image creatives with ≥90% classification accuracy.

11 Judge Quick-Start & Submission Details

Step-by-step instructions for testing the live system and complete submission metadata

Quick-Start Judge Evaluation Instructions

1. **Open the Live Web Application:** Access https://brandsentry-web.onrender.com in any modern desktop browser.
2. **Ingest Guideline:** If the database is empty due to a Render free-tier cold restart, click the *"Load Sample Marketing Compliance Guidelines"* button, or upload any PDF/TXT guideline.
3. **Test Supported Claim:** Enter `"Our marketing materials must include traceable evidence sources for every material factual claim."` → Verify green `SUPPORTED` verdict and exact citation quote.
4. **Test Prohibited Guarantee:** Enter `"Our product is 100% guaranteed to deliver perfect results."` → Verify red `CONTRADICTED` verdict and suggested rewrite.
5. **Test Missing Evidence:** Enter `"Our product increases customer productivity by 85%."` → Verify amber `INSUFFICIENT_EVIDENCE` verdict and zero fabricated citations.
6. **Test Semantic Drift:** Enter `"Our product always improves productivity."` → Observe the Semantic Drift panel highlight `CERTAINTY_SHIFT`.

Official Submission Endpoints & Links

Resource Name	Direct Endpoint / URL	Verification Notes
Live Web Application	`https://brandsentry-web.onrender.com`	OPERATIONAL
Backend REST API	`https://brandsentry-api.onrender.com`	OPERATIONAL
Interactive API Docs	`https://brandsentry-api.onrender.com/docs`	FASTAPI SWAGGER
Backend Health Check	`https://brandsentry-api.onrender.com/health`	HTTP 200 OK
Source Code Repository	`https://github.com/KPCHIRAGGUPTHA/BRAND_SENTRY_AI`	PUBLIC GITHUB

Participant & Team Details

Team Name: Nexora AI
Participant Full Name: K P Chirag Guptha
Registration Number: 24MIS0369
Institution: Vellore Institute of Technology (VIT)
Participation Category: Individual Solo Participant

Closing Engineering Statement

BrandSentry AI has successfully evolved from a conceptual proposal into a verified, publicly accessible compliance review system. Built with deterministic safeguards, zero hallucinated approvals, and semantic drift detection, it delivers dependable decision support for marketing teams.